buildings

MDPI

Article

Exploring Gamification's Role in Shaping Socially Sustainable Urban Spaces: A Case Study of Gensen in SOLANA, Beijing

Yanhua Yao [1,*], Zheyu Li [2] and Sai Ma [1]

[1] Academy of Arts & Design, Tsinghua University, Beijing 100084, China; masai@mail.tsinghua.edu.cn
[2] Prismo Studio, Shanghai 200030, China; lizheyu@prismostudio.cn
* Correspondence: yyh19@tsinghua.org.cn

Abstract: This paper explores the innovative integration of digitalization and gamification in urban design to address social sustainability challenges in rapidly evolving cities. Using Gensen, a metaverse platform launched in 2024, as a case study, the research investigates how the convergence of virtual environments and real urban spaces can foster new forms of social interaction, spatial usage, and community engagement. The motivation behind this study is to assess the potential of gamification in enhancing social sustainability within digitalized urban contexts. By introducing the opening event, "Treasure Hunting", established by Gensen, the study examines how users engage with existing urban settings through a gamified approach. The research investigates how gameful design, facilitated by digitalization, can transform public spaces into more dynamic environments that encourage diverse participation while also addressing emerging risks such as spatial inequality resulting from varying levels of access to digital tools and literacy. The research questions focus on how gamification can bridge these gaps and contribute to creating more inclusive urban environments. In conclusion, the study argues that current gamified design approaches, based on digitalization, often overlook the playfulness inherent in physical environments and human interaction. However, a bottom-up approach that emphasizes individuals' understanding of the inherent playfulness in existing urban spaces is still lacking. This aspect needs further exploration to inform and enhance gameful design strategies aimed at promoting social sustainability in urban development. Bridging this gap is essential for integrating digital interventions into everyday life, ultimately achieving a more effective gameful design in urban contexts.

Keywords: gameful design approach; digitalized lifestyle; gamification; social sustainability

check for updates

Academic Editors: Nikos A. Salingaros, Aso Hajirasouli, Ehsan Noroozinejad, Gregory M. Morrison, Mike Kagioglou and Nashwan Dawood

Received: 28 March 2025
Revised: 27 May 2025
Accepted: 4 June 2025
Published: 6 June 2025

Citation: Yao, Y.; Li, Z.; Ma, S. Exploring Gamification's Role in Shaping Socially Sustainable Urban Spaces: A Case Study of Gensen in SOLANA, Beijing. *Buildings* **2025**, *15*, 1969. https://doi.org/10.3390/buildings15121969

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Cities and games are intricately linked. Real-life cities often serve as core references in the design of game levels, using the cultural and architectural styles of cities as essential tools to enhance player attachment and engagement. Additionally, individual behaviors within and beyond the game environment are governed by principles that also apply to urban spaces. This connection highlights a shared underlying dynamic rooted in individual subjectivity, influencing interactions in both virtual environments and real-world urban settings. As virtual and real spaces increasingly integrate through advancing digital technology, this presents a novel approach to understanding environmental behavior within a virtual-real world context. Such integration opens opportunities for the establishment of new socio-spatial cultures that enhance social cohesion through a co-design approach, ultimately contributing to more sustainable urban development.

The social dimension is fundamental to sustainability. Despite the extensive body of definitions and frameworks developed across various fields, the concept of social sustainability remains ambiguous when compared to environmental or economic sustainability [1,2]. Contemporary approaches to social sustainability converge around concerns of equity, identity, participation, and the lived experience of urban life [3–5]. The urban context serves both as a lens through which social sustainability is perceived and as a complex factor in its development, shaping how it is implemented and understood. The incorporation of aspects of individuals' everyday lives blurs the boundary between the natural and built environments [6], prompting a reevaluation of the traditional Venn diagram structure of sustainability, which typically encompasses the environmental, economic, and social dimensions. A more holistic approach, rooted in individuals' everyday practices, offers a redefined concept of sustainability, suggesting that it should not be understood as separate components. This perspective is further supported by advancements in digital technology, which facilitate more integrated and dynamic approaches to sustainability in urban environments.

Urban renewal, regeneration, and design processes illustrate how social sustainability can be integrated into the development of people's living and working environments. These processes include aspects such as housing, infrastructure, urban form, greenery, accessibility, the quality of public spaces, and community [2,7]. More importantly, the allocation and distribution of resources play a critical role in shaping the relationship between space and people, enabling safe, interactive, and easily accessible environments [8]. The physical form of cities is a vital instrument for achieving social sustainability by facilitating interactions with individuals' everyday behaviors. These forms are transformed by people's daily activities, which are increasingly influenced by digitalization. Additionally, social life has gradually shifted from physical spaces to digital realms, especially in urban settings and mobility [9]. This shift has profound implications for how social sustainability is defined and realized, as the digital realm now plays a central role in shaping social interactions, access to services, and mobility patterns within cities. Therefore, further design attention from a political perspective—such as supportive policies, participatory planning, and inclusive design frameworks—is required to treat the integration of virtual and physical spaces as a foundational condition for socially sustainable urban development.

Advances in urban sensing technologies have led to the generation of vast datasets, providing valuable insights into how individuals interact with urban environments [10]. These datasets, collected from everyday urban life at high resolution, empower Artificial Intelligence (AI) to play a pivotal role in sustainable urban design and planning [11,12]. While planning decisions at the macro scale, supported by computer algorithms, have significantly advanced environmental and economic sustainability, they do not fully address the complexities of social sustainability. While the complexity of social sustainability continues to grow—both in terms of its internal dimensions (i.e., the integration of its three core components: social equity, community development, and quality of life) and its applied context (i.e., the integration of digital and physical spaces)—a research gap emerges in understanding how advanced technologies can effectively address this dual-layered complexity. Specifically, there is limited investigation into how technological innovations, such as gamification and AI-driven design, can contribute to social sustainability in urban environments by bridging these intertwined dimensions.

This paper reports on data collection from a pilot urban digitalization and gamification project in Beijing, providing a case study to explore how gamification might enable social sustainability by impacting individual behaviors. More importantly, it critically discusses the interplay between the gamification of urban space promoted by the application and the gamefulness and playfulness already embedded within the city. Gensen, launched in 2024,

is a pioneering mobile application that integrates urban space with virtual environments, offering a novel approach to examining the dynamic interrelationship between the two. The collected data has led to the development of an innovative urban design approach that fosters interaction in public spaces by actively involving individuals in the process of shaping urban environments. Check-in data was gathered during the initial event—a "treasure hunt" activity held over four days, beginning on 20 July 2024. The event demonstrated how the digitalization and gamification of urban space can enhance public space accessibility and cultivate cultural communities, distinct from traditional spatial community interdependence mechanisms. This study seeks to examine how gamification can serve as an effective tool in urban design within increasingly digitalized living environments to support social sustainability. More specifically, it investigates whether digital game-like applications can fulfill the aims of gameful design by enhancing citizen engagement with urban space and daily life. Preliminary data mapping and analysis suggest the potential of this approach to inform a co-design strategy for the built environment. This strategy could provide valuable insights into how gamified urban spaces might be designed in the future, facilitating the creation of vibrant, adaptive communities.

2. A City with Gamefulness

2.1. The Constitution of Gameful Design/Gamification

Gamification has become a crucial tool in various sectors, including finance, health, education, sustainability, and community engagement [13]. It applies game-like experiences, driven by motivational affordances, to enhance services and engage users effectively. Gamification comprises three core components: motivational affordances, the resulting psychological outcomes, and the subsequent behavioral outcomes [14]. Gamification refers to the use of game elements, features, and design in non-game environments, contexts, or services [15–17]. This differs from playful design, which focuses on creating enjoyable interactions without necessarily incorporating the full range of game mechanics [18].

Gameful design, or "gamefulness", aims to create game-like experiences by utilizing game elements. This approach is not limited to digital contexts but can also be applied across various platforms and mediums. In Huotari and Hamari's [19] view, games can be conceptualized as systems requiring active participation from players, with both developers and players acting as co-producers of the experience, making the game unique and subjectively experienced.

Deterding [18] further explains that the gameful design elements identified by Reeves and Read are suitable for contexts outside of traditional game environments. These elements include self-representation with avatars, 3D environments, narrative context, feedback, reputation, ranks, levels, marketplaces and economies, competition under explicit rules, teams, parallel communication systems, and time pressure. Tondell [20] identified 49 elements, both within and outside academic fields, and Tondello and Nacke [21] classified these elements into several categories: socialization, assistance, immersion, risk/reward, customization, progression, altruism, and incentive.

These gamification elements are not confined to digital game environments but are increasingly embedded in everyday life. They function as instruments that influence psychological states and behavior, shaping individuals' experiences in both physical and virtual spaces. Elements like feedback, progression, and self-representation help construct identity, motivation, and engagement, ultimately influencing how individuals learn and interact with their environments [22,23], mediating perceptions and interactions across both physical and virtual domains. Elements such as feedback, progression, and self-representation contribute to the construction of identity, motivation, and engagement, thereby influencing how individuals learn, experience, and navigate their environments [24–26]. These

elements not only influence interactions with the material environment but also actively contribute to the achievement of sustainable development goals through changes in individual behavior [27].

Socialization, in particular, serves as a central mechanism linking digital and physical spaces. As a shared platform, it fosters relational dynamics and collective experiences, emphasizing the convergence of virtual and real-world interactions. By embedding gameful design elements in daily contexts, social sustainability is promoted by encouraging participation, community cohesion, and a heightened sense of agency.

2.2. Gamefulness in Digitalized Urban Life

Digitalization and data play a pivotal role in the development of smart cities, aiming to improve quality of life through enhanced safety, reduced crime, lower disease rates, decreased commute times, and overall urban efficiency. This approach has been widely adopted by policymakers globally [28]. Smart infrastructure in cities, including sensing technologies, artificial intelligence (AI), and the Internet of Things (IoT), is deployed across various facets of daily life, forming the technological backbone that shapes human practices. These technologies not only enhance quality of life but also significantly impact urban governance, both from behavioral and social perspectives [29]. The interface between individuals and urban space has been digitalized, enabling a transformation in urban design mechanisms for social sustainability. This shift moves the focus from urban forms to urban living behaviors. However, a gap in access to smart technology persists, particularly concerning age, education, and rural versus urban divides [30]. This digital divide could exacerbate disparities, negatively impacting social cohesion and altering work structures. Despite the widespread accessibility of digital technology, these gaps still present a challenge, particularly in how advanced living environments and digital tools, including gamification, are utilized. Furthermore, the cultural and historical context of cities is critical when establishing the future of smart cities. Many smart city initiatives tend to overlook the deeply rooted local cultures, traditions, and histories that shape the identity of urban spaces. This neglect can result in a disconnection between technological advancements and the lived experience of local communities. As a result, smart cities may fail to resonate with their inhabitants, particularly when the designs imposed by these technologies do not align with or respect local cultural practices and histories [31]. In this context, the integration of gameful design approaches can serve as a crucial tool in bridging the gap between digital innovation and social sustainability. Gameful design, by encouraging interaction, collaboration, and engagement, enhances the "smartness" of a city not just in terms of technology but also through the active participation of its residents.

Firstly, integrating virtual games with real-world urban spaces represents a powerful intersection of technology, design, and human interaction. Well-known examples like Pokémon GO and GTA Online illustrate how virtual game environments can act as digital extensions of urban spaces, intertwining the real-life experience of the city with the virtual objectives of users. Nevertheless, non-commercial platforms such as Foursquare, FixMyStreet, and Playable City also demonstrate the growing potential of gameful design in civic and urban contexts. These platforms contribute to the integration of virtual and physical environments by fostering participatory engagement, enhancing place-awareness, and encouraging playful interactions with urban infrastructure. New forms of sociability and freedom have emerged through engagement with simulated urban environments, mediating and extending the meaning and experience of real urban life. As interactive social worlds, games influence and reshape our understanding of urban experiences through their designated virtual spaces [32]. Secondly, as a digitalization tool for urban space, games facilitate the formation of social networks online, broadening the role of games

as vital components of leisure and entertainment in daily activities. Online communities within large virtual worlds are created through games [33], which play a significant role in establishing cultural and social connections [34,35]. With online communities becoming increasingly important for leisure activities [36], the merging of virtual and urban spaces enhances the dynamics of daily life. Playfulness is essential in both online and offline spatial design, acting as a bridge between the two through individual engagement.

Gamification, when applied to urban spaces, is not merely about overlaying a digital game layer onto existing urban fabric. Rather, it seeks to embed gameful design into increasingly digitalized lifestyles—including activities like shopping, dining, and working. Over the past decade, e-commerce has rapidly accelerated globally, with gamification emerging as a key strategy for enhancing user engagement and influencing consumer behavior [37–39]. Recent studies highlight the growing relevance of gamification as a strategic tool in enhancing user engagement and influencing consumer behavior in e-commerce environments. Gamification has demonstrated positive effects on key customer outcomes such as brand engagement, purchase intention, customer loyalty, and retention [40,41]. Gamification has emerged as a promising approach in food-related technologies, aiming to influence user behavior in areas such as sustainable consumption, healthy eating, and labor engagement [42]. In online food delivery services, gamification is used to increase worker motivation and engagement, addressing challenges related to platform management and commitment [43]. Additionally, gameful design in food-related technologies, particularly in physical retail environments, has been shown to enhance users' food literacy, promote healthier food choices, and reduce impulsive behavior [44]. Across these domains, gamification demonstrates potential to reshape interactions with food systems by fostering motivation, supporting informed decision-making, and encouraging long-term behavioral change.

Digitalization and gamification have thus become pervasive in modern urban life, blurring the boundaries between work and play, design and behavior, and the real and simulated. As digital platforms evolve and gain traction, the intersection of gaming and virtual reality has emerged as an integral part of the Metaverse, further enhancing the gaming experience [45]. The widespread adoption of gameful design in cities suggests that gamification has evolved into a cultural logic embedded in the infrastructures of urban life. This process is redefining how urban spaces are experienced, facilitating a dynamic and interactive engagement between individuals and their environments.

2.3. Gamified Urban Design as a Strategy for Social Sustainability

The concept of "play" has long been embedded in urban spaces, not only through traditional game approaches [46] but also as a mechanism for developing typologies in urban design interventions [47]. The "playful city" concept integrates play and playfulness into urban environments to promote social interaction, spatial exploration, creativity, and engagement with public spaces [48]. The relationship between play and urban life is inherently multifaceted. Historically, cities and their architectural aesthetics have been conceived as works of art that create space for play and entertainment—realms consciously positioned outside the spheres of labor, politics, and utility. This separation allowed urban public spaces to be transformed into spectacles of leisure [49].

Gameful design in the city, on the other hand, presents a different trajectory from these historical notions of passive entertainment. As Alfrink [50] critiques, serious games and gamification often reduce play to a utilitarian activity, stripping it of its cultural and creative richness. Unlike the spectacle-driven environments of the past, gameful design encourages an open, adaptive, and participatory urban experience. It is not about prescribing how people should play or live but enabling conditions for appropriation and co-creation by

urban inhabitants. As Sanoff [51] suggests, games provide a mechanism for engaging participants in learning processes by enabling them to explore values, ideas, and behaviors through role-playing and interactive decision-making. This participatory approach helps individuals better understand both themselves and others, offering insights into complex social dynamics that may otherwise be overlooked. Nevertheless, playful city is not simply a setting adorned with games or interactive technologies. It is a conceptual and material landscape that recognizes play as a foundational mode of urban engagement [52]. Three core premises guide playful and gameful design in urban spaces: (1) play and games can unlock citizen intelligence by decentralizing it in urban contexts; (2) games can organize engagement and collective agency, fostering new forms of citizen participation through shared goals, temporary communities, and spaces for negotiation; (3) gameful design is not about controlling behavior through gamification but offering open-ended frameworks that allow residents to explore, inhabit spaces, and form relationships.

Recent scholarship emphasizes the integration of gamification into urban planning and sustainability. For example, Ampatzidou [53] illustrates how iterative game design—exemplified by the Sustainability Futures Game—supports long-term transformation by enabling collective visioning aligned with the UN Sustainable Development Goals. It emphasizes the value of openness in design processes: openness in research intent, thematic scope, and target audiences. While this flexibility fosters creativity and broad applicability, it also presents challenges in translating game outputs into concrete planning actions. Similarly, Angelidou and Psaltoglou [54] find that gaming approaches such as Block by Block and CityScope help "decode" urban processes, enhancing transparency and inclusivity, although the level of participant empowerment varies. Ferri [55] proposes playful urban artifacts as alternatives to technocratic smart city models, arguing that ludic interaction can communicate sustainability goals in non-instrumental, participatory, and culturally expressive ways. Building on this, Fox [56] advocates for gamified decision-support systems (DSSs) that combine climate education with participatory greenspace planning, enabling more socially and environmentally responsive outcomes in climate-vulnerable urban contexts.

Despite the potential for gamification to enhance citizen engagement and foster inclusive urban futures, a critical gap remains. The systematic integration of gamification into urban design practices is still underdeveloped. While research in the service industry and game design has explored the social sustainability implications of gamified interventions in public spaces, this area remains under-theorized. Furthermore, traditional urban design approaches have often focused on physical forms and spatial configurations while overlooking behavioral dynamics and experiential dimensions. Therefore, further research is needed to bridge these gaps, moving beyond isolated applications and toward sustained, scalable practices that contribute to socially sustainable urban transformation. This study aims to analyze how gameful design elements are deployed in existing platforms and the extent to which they influence people's behavior and movement within urban spaces.

3. Method and Research Subject

3.1. Method

This study employs the recently launched platform Gensen (version 1.0.8) as a case study to examine how the integration of digitalized urban spaces and gamification—based on existing urban settings—can contribute to enhancing social sustainability. By critically analyzing the gameful design approach adopted by Gensen in its application of urban space, the research explores how such methods might inform future urban development in the context of advancing digital technologies. The methodology follows a design- and behavior-

oriented inquiry connecting theoretical frameworks, spatial data, and participatory design analysis (Figure 1).

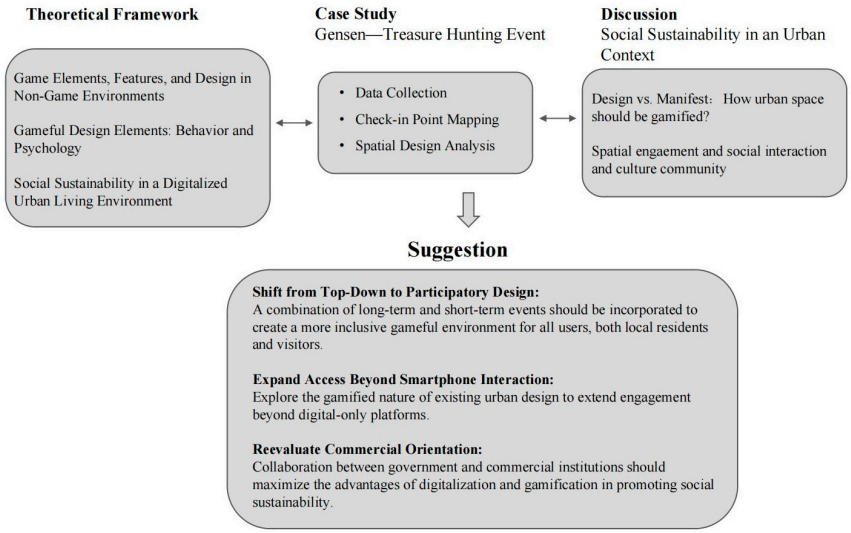


Figure 1. By applying the case study of Gensen, the study aims to bridge the theoretical framework and potential solutions for improving social sustainability through a gameful design approach, while also further contributing relevant suggestions for future development.

3.1.1. Background

The rapid advancement of digital development in Chinese cities has significantly enhanced citizens' quality of life [57]. This transformation not only influences how individuals conduct their daily activities but, more importantly, it reshapes the way they interact with the material urban environment. In this evolving digital landscape, the Chaoyang District Council launched the Three-Year Action Plan for New Development in Internet 3.0 (2023–2025), which aims to expand the application of advanced digital technologies to support commercial vitality while simultaneously improving residents' everyday experiences. At the municipal level, the Beijing Territorial and Spatial Plan (2021–2025) reinforces this vision by promoting the regeneration of the city's central area to foster urban vitality and realize the value of small-scale, human-centered characteristics [58]. These strategic policies are converging to drive a new wave of digital urbanization in Beijing, creating a unique context in which digital infrastructures, spatial planning, and commercial development are increasingly interconnected. Within this framework, pilot projects like Gensen have emerged as experimental platforms through which these digitalization strategies are deployed and tested.

Gensen is a metaverse platform launched in June 2024, designed by Prismo Studio, that operates on mobile phones and allows users to interact digitally. Gensen provides a medium for users to socialize leisurely and shop, which engages with massive behavioral and check-in data generated through the digitalization and gamification of urban spaces. The platform's pilot implementation occurred at the SOLANA commercial retail zone, located within the northeastern 4th ring road in Chaoyang District, Beijing (Figure 2a). This central shopping area opened in 2008—the same year as the Olympic Games—and has seen significant growth in its annual passenger flow volume, increasing from 9.05 million in 2010 to 15.5 million in 2020. The average number of visitors reached 370,000 on weekdays and 560,000 on weekends. SOLANA serves as a typical urban public space, combining multiple functions with over 500 shops that offer shopping, education, entertainment, culture, leisure, and dining, forming a combined outdoor and indoor space. Gensen aims to build a virtual reflection of SOLANA, including the skyline, landscape, street furniture, retail stores, and greenery. It also extends the visual appearance for different events and

scenarios of the place accordingly (Figure 2b). This platform exemplifies the digitalization of urban spaces based on integrating consumption and gaming within the Chinese context. Within 7 days after the launch of the project, the registered users have reached 7700, with an average of 380 active users every day.



(**a**)



(**b**)

Figure 2. Pilot urban digitalization project at SOLANA, Beijing (**a**) supported by Gensen, Prismo Studio. A virtual replica of the real-world space is built within the application to create a more immersive user experience (**b**).

A modular prototype has been implemented to model both outdoor public spaces and retail interiors, serving the diverse needs of businesses that operate at the intersection of offline retail and online consumption. This prototype enables rapid updates related to promotional events, new product releases, and spatial design changes, while maintaining seamless integration with the 3D virtual representation of SOLANA (see Figure 3). The rendering process is driven by predefined keywords linked to a shared 3D model framework, minimizing the need for large-scale content downloads and enhancing performance efficiency.

A semi-generative spatial design approach underpins this system, establishing a flexible framework that allows brands and retailers to design their own virtual environments. By incorporating keyword-based data collection from users—including visitors, local residents, business owners, and designers—this approach fosters participatory design and collaborative spatial thinking. These collected inputs can be further used in future stages through GAN (Generative Adversarial Network)-supported spatial design applications for both virtual and physical contexts. A modular game system is embedded within this 3D urban environment, aligning with the modular architecture of the overall design. Users interact with gamified features to earn discounts and cash vouchers, which can be redeemed at physical retail locations. Central to the functionality of this gamified ecosystem is a robust electronic payment system that enables the seamless conversion of virtual rewards into tangible goods and services. The success of this system relies on coordination among multiple stakeholders—including SOLANA's management entity, partner retail outlets, and digital payment service providers—to ensure the smooth redemption of user-earned credits across key domains such as shopping, parking, and service top-ups.

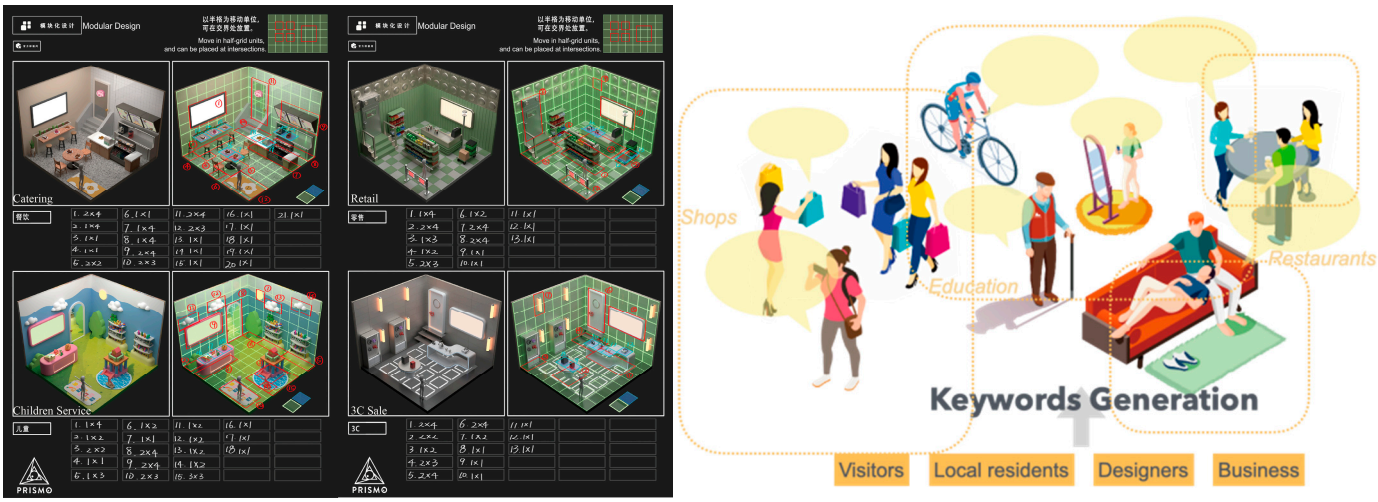


Figure 3. A semi-generative 3D interior design model for different types of stores in SONALA based on keywords generated by the public.

3.1.2. The Treasure Hunting Game

One of the key strategies for gamifying urban space and everyday life—emerging from the collaboration between Gensen and SOLANA—is the continuous organization of hybrid events that bridge virtual and physical environments. Retail, gaming, social interaction, and payment comprise the four primary modules that structure the virtual urban experience of SOLANA, mirroring visitors' daily behaviors (see Figure 4). The events, such as art exhibitions, festival celebrations, and pop-up brand installations, which were originally launched frequently by SOLANA, will now be incorporated into Gensen. These events will function as critical interfaces between the digital and built environments, encouraging users to engage both virtually and physically. To participate in designated activities and games, users are required to be physically present at SOLANA, where they can earn redeemable digital vouchers usable within the local area and across associated brand stores.



(**a**) (**b**)

Figure 4. Real-life scenes built in Gensen, such as purchasing at a café (**a**) and urban gardening (**b**), integrate real-life and gaming experiences. Both permanent and temporary events are consolidated into a single platform, forming the foundation of a gameful design approach.

The Treasure Hunting game was designed as one of the opening events of the Gensen pilot, serving both promotional and evaluative purposes. As a launch activity, it aimed to generate public interest and engagement with the application while simultaneously testing the software's functionality and usability. This event provided a semi-controlled setting in which user interactions with the digital platform could be observed in a real-world urban context, offering valuable insights for the iterative refinement and future development of the Gensen system.

The event was designed to promote user engagement by encouraging participants to check in at both online and offline retail locations. To complete a check-in on the Gensen platform, users were required to be physically present within a 50 m radius of the designated shop. A total of 27 shops were selected for the event, each identified as a location to be promoted and explored (Figure 5). Participants who successfully checked in at a minimum of six different designated shops were rewarded with a voucher redeemable for a complimentary cup of coffee. This incentive structure aimed to stimulate movement across the site and enhance user interaction with both the digital platform and the physical commercial environment. While the existing strategy for public participation on Gensen primarily considers younger generations—who represent the platform's main user base—individuals with relatively low digital literacy may be underserved. Prior knowledge of the area, such as familiarity with the layout, shop locations, and distances between shops, may also differ significantly between local residents and visitors.



(**a**) (**b**)

(**c**) (**d**)

Figure 5. The application interface of Gensen for check-in at designated stores (**a**), where users can click the logo to enter the store by walking within 50 m of the store (**b**). After entering, users can click the "Store Check-in" button (**c**). After successfully checking into six stores, users can earn a coffee voucher (**d**).

To address these concerns, future iterations of Gensen's gamified approach should account for the needs of diverse population groups, including older adults, people with disabilities, and those less familiar with digital tools. This can be achieved through a more inclusive design strategy that provides a user-friendly interface adaptable to a broad range of users, regardless of their age, ability, or spatial knowledge of the site. By minimizing potential participation bias, such improvements would contribute to achieving the project's goal of enhancing social sustainability on multiple levels.

3.2. Data and Analysis

Data for this study was generated through user check-ins during the Treasure Hunting event, where participants were required to locate designated shops—referred to as "treasures"—as well as through coffee voucher redemptions. The raw dataset includes attributes such as user ID (anonymized), check-in date and time, shop ID, shop name, and the geo-location of each check-in. While demographic characteristics of users were

not accessible due to privacy policies, the available data allows for spatial and temporal analysis of user behavior within the SOLANA commercial complex.

The primary objective of the data analysis is to explore how Gensen may facilitate user interaction with and exploration of the physical environment, particularly in areas that previously experienced lower levels of foot traffic or accessibility. However, the absence of user demographic data—such as gender, age, or occupation—limits the ability to assess differentiated user experiences or to draw conclusions about inclusivity and behavioral variance across population groups. This lack of granularity restricts the potential for understanding broader patterns of social equity or targeted user engagement. To address this limitation, future studies could incorporate the collection of user demographic data, with users' informed consent, during platform registration or interaction. This would not only strengthen the foundation for further development of the application but also enable integration with broader research domains, such as urban health, safety, and mobility behavior at scale. Additionally, the use of proxy indicators—such as device type, check-in time, or travel frequency—could offer preliminary insights into user profiles and lifestyle patterns. Enhanced data, including high-resolution movement trajectories, general home location, and potential integration with other social media platforms, could further support the construction of a more comprehensive understanding of digitally mediated urban social networks.

Additionally, potential biases may arise due to participants' prior familiarity with the SOLANA space and their experience using the mobile application. Regular visitors to SOLANA may already be more comfortable navigating the environment, which could influence their engagement levels during the event. Similarly, participants who are more familiar with mobile applications and gamified systems might exhibit different interaction patterns than those less experienced with digital tools. These factors could influence the data and should be considered when interpreting the results.

Nonetheless, the purpose of this study is not to evaluate the precise social impact of the Gensen platform at the individual level but rather to examine how its gamified approach aligns with, diverges from, or potentially extends established principles of gameful design in urban contexts. By analyzing spatial and temporal user interactions in the pilot event, the study seeks to assess the operational potential of gamification as a systemic design strategy.

4. Preliminary Analysis

This paper presents data gathered from 20 to 23 July to illustrate how a designated event can attract people to the study area on top of its original design. During this four-day trial, check-in locations from 88 users were collected, resulting in 272 trips, and all data was acquired for this analysis. A total of 73 users gained the voucher during the event, which is about 83% of total users. Most users spent 3 or 4 days completing the task, while 10 users finished the 6 check-ins within 1 day and 7 users within 2 days. However, users with fewer attending days tended to fail the task.

The designated check-in shops were primarily concentrated in the central area of SOLANA, where three distinct spatial clusters emerged—around the Central Plaza, the Green Pathway, and the indoor shopping area in the eastern section. Two of these—namely, the Central Plaza and the Green Pathway—are key landscape features that naturally attract visitors due to their relatively high levels of visual accessibility. These areas contribute to a dynamic and engaging spatial experience across the site (Figure 6). In addition to spatial data, the study incorporates user-generated content from Dianping, one of China's most widely used review and recommendation platforms. Photographs and accompanying comments shared by users further emphasize the significance of these two focal areas,

which consistently appear as preferred locations for photography and public reflection on the SOLANA experience (Figure 7).

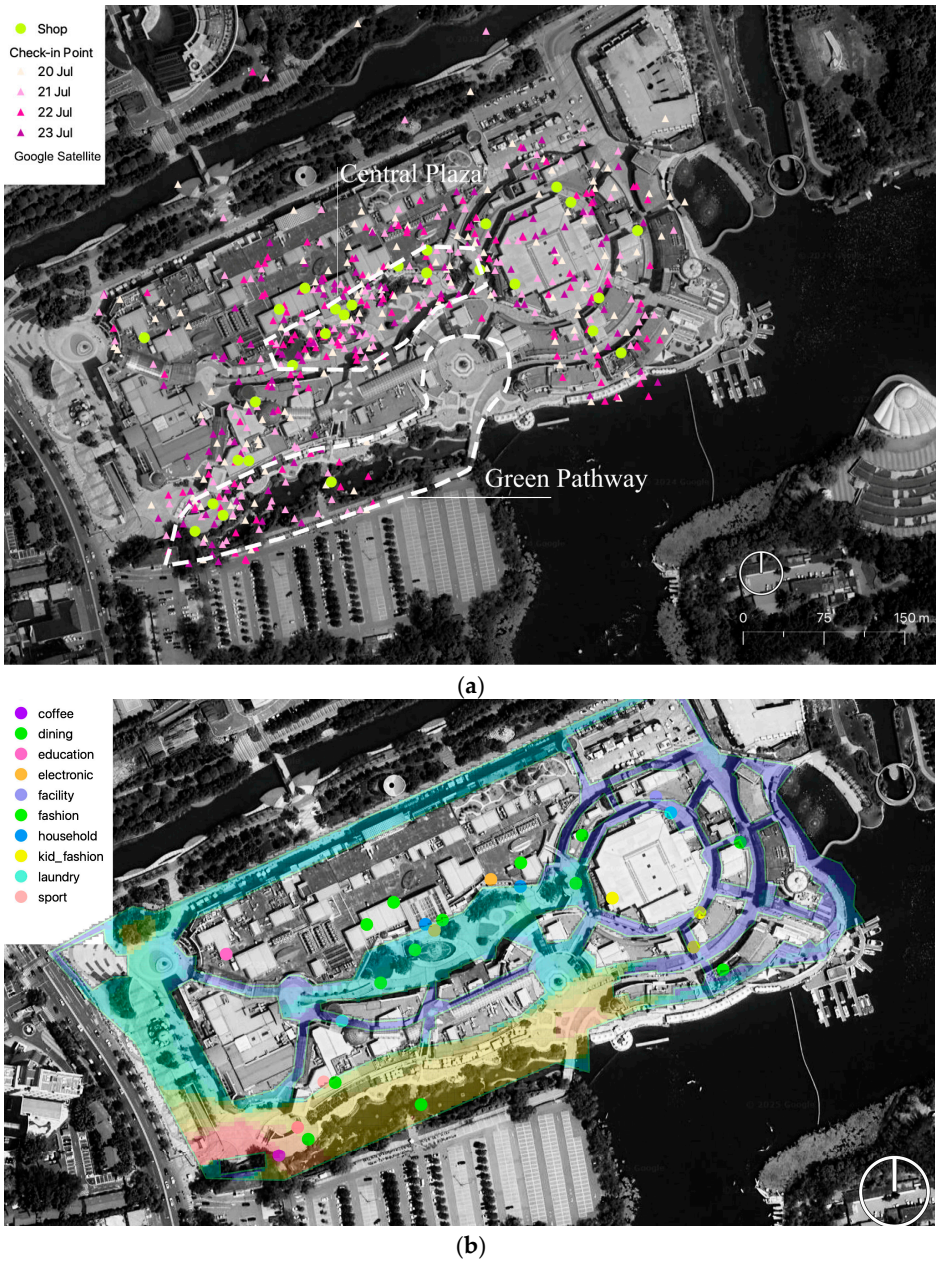


Figure 6. Two primary landscape attractions in SOLANA serve as proximity anchors: (**a**) the mixed-use area at the center and the dining area in the southern section. These areas exhibit high levels of visual accessibility, particularly when compared to the eastern indoor shopping zones, which are less visually exposed (**b**). While players were naturally drawn to these visually prominent areas, the design of the Treasure Hunting game strategically encouraged exploration of less frequented or harder-to-find spaces within SOLANA, enhancing spatial engagement across the entire site.

However, analysis of check-in data collected during the 4-day Treasure Hunting event reveals a more nuanced spatial pattern. While user activity does partially overlap with these visually accessible attractions, check-in frequency is more strongly aligned with the location of the designated shops—some of which are situated in areas not typically associated with high foot traffic. Notably, the eastern portion of SOLANA, characterized by indoor retail spaces and a relative lack of multifunctional features, tends to be underutilized under normal conditions. Yet, during the gamified event, this area saw a significant increase

in visitor activity. A similar trend was observed at the northeastern entrance—another area that typically receives limited attention. These findings suggest that the game-based structure of the event effectively motivated participants to navigate beyond familiar or visually prominent zones, thereby enhancing spatial inclusivity and activating underused areas within the urban commercial landscape.

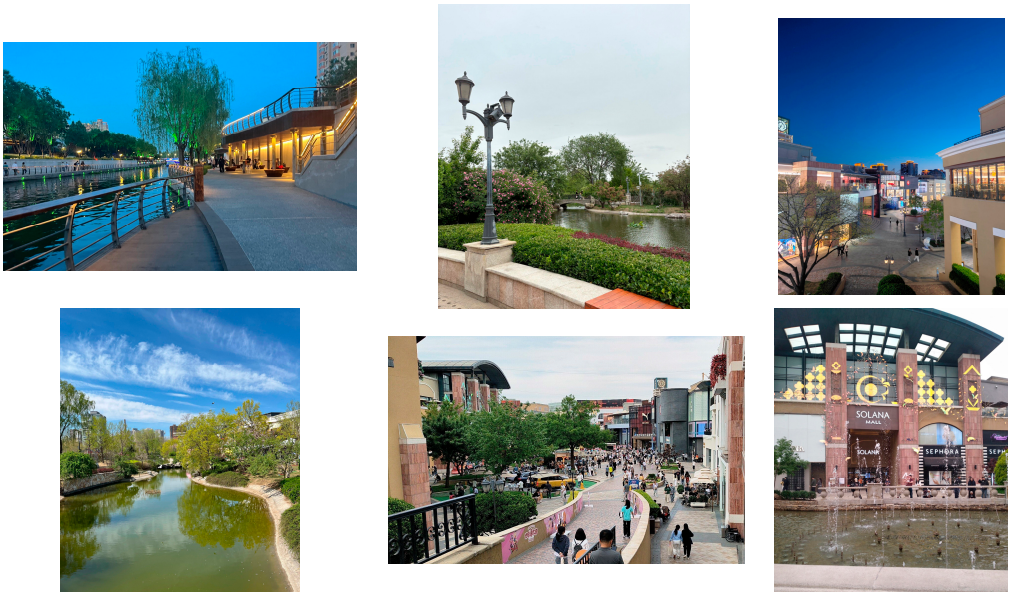
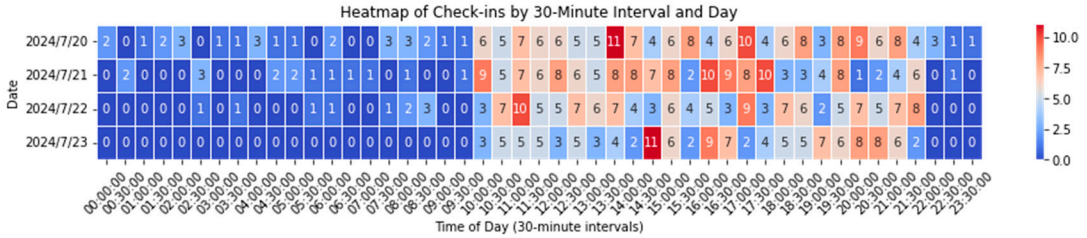


Figure 7. User-generated images of SOLANA's Central Plaza and Green Pathway, uploaded to Dianping, illustrating these areas as prominent sites of visitor interest and spatial engagement.

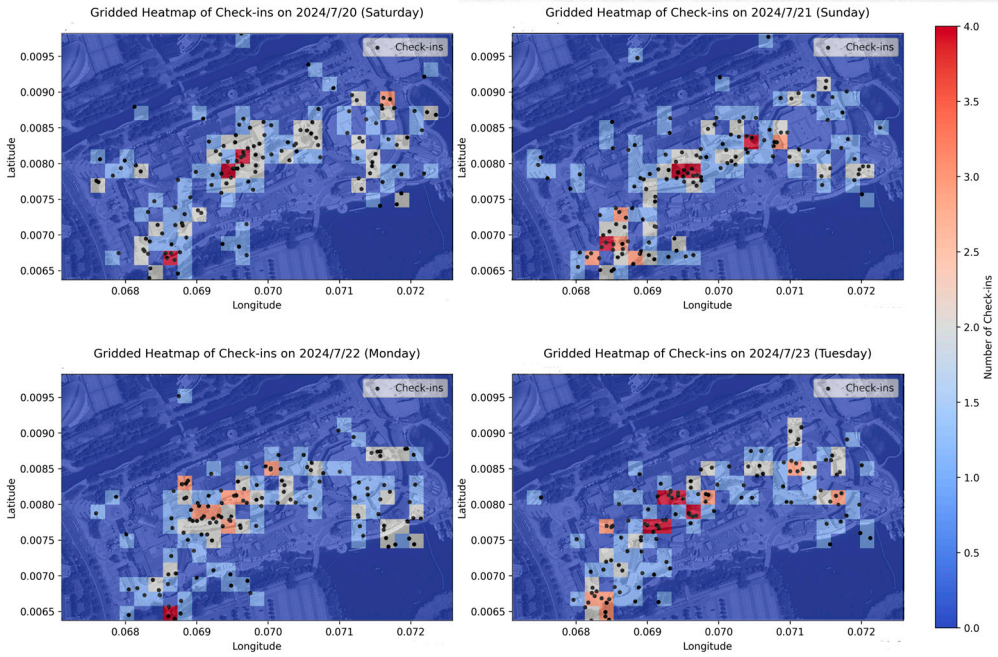
Moreover, among the 26 designated shops, approximately 35% of the check-ins occurred at fashion stores, while 27% were at restaurants and cafés. To conduct the spatial-temporal analysis of the check-in data, we used raw information containing the timestamp and geo-coordinates (longitude and latitude) of each user interaction. The check-in locations were categorized by store type, and spatial distributions were generated by mapping these points. We then examined temporal and spatial patterns by comparing check-in frequency across different times of day and between weekdays and weekends. Heatmaps were produced to visualize peaks in user activity and reveal spatial clustering. The results indicate notable variations in peak hours and locations between weekends and weekdays (Figure 8), reflecting diverse patterns of user engagement. These areas functioned as temporary destinations within SOLANA, contributing to more dynamic and distributed patterns of urban mobility.

By comparing the types of activities encouraged by the existing spatial configuration of SOLANA, the gamified design approach used in this study appeared to improve accessibility to commercial areas. For instance, it helped draw attention to less-visited shops and encouraged weekday use of restaurants and cafés. However, while these patterns reflect increased user mobility, they do not necessarily translate into improved social sustainability in terms of social mixing or community building—particularly in a commercially driven development context, even when supported by a policy-oriented platform like Gensen.

Although Gensen aims to foster interaction across different user groups, the current implementation—primarily a location-based, goal-oriented experience—resembles a single-player model. Such game structures tend to limit opportunities for group interaction or a sense of shared community. This highlights a critical limitation of gamification in urban design: without deliberate integration of collective or cooperative tasks, these systems may reinforce individualistic behaviors and overlook the potential for fostering social cohesion.



(**a**)



(**b**)

Figure 8. The temporal and spatial distribution and concentration of check-ins. The check-in times were concentrated between 09:30 and around 21:30 for all four days. A relatively higher number of check-ins occurred during the weekend, while Monday and Tuesday, which are usually less frequently visited, remained active (**a**). Spatially, however, the concentration of visits still highly corresponds to the Central Plaza and Green Pathway (**b**).

To better support social sustainability, future gamified design approaches should be adapted to their specific spatial, cultural, and community contexts. Game mechanics should be designed to encourage not only movement and exploration but also inclusive participation and interpersonal engagement. Without such considerations, gamification risks unintentionally reinforcing social exclusion, particularly among populations with lower levels of digital literacy or limited access to mobile technology.

5. Discussion

5.1. A Gamified Urban Design Approach

The Treasure Hunting event in SOLANA illustrates how gamification can temporarily reconfigure the spatial dynamics of urban environments by layering additional meaning onto existing public spaces. The clustering of user check-ins corresponds to designated shop locations and game rules, reflecting how digital prompts can influence mobility patterns and spatial interaction (Figure 9). This suggests a transformation of the urban landscape into a semi-structured game environment, where routine movement is redirected through playful engagement.

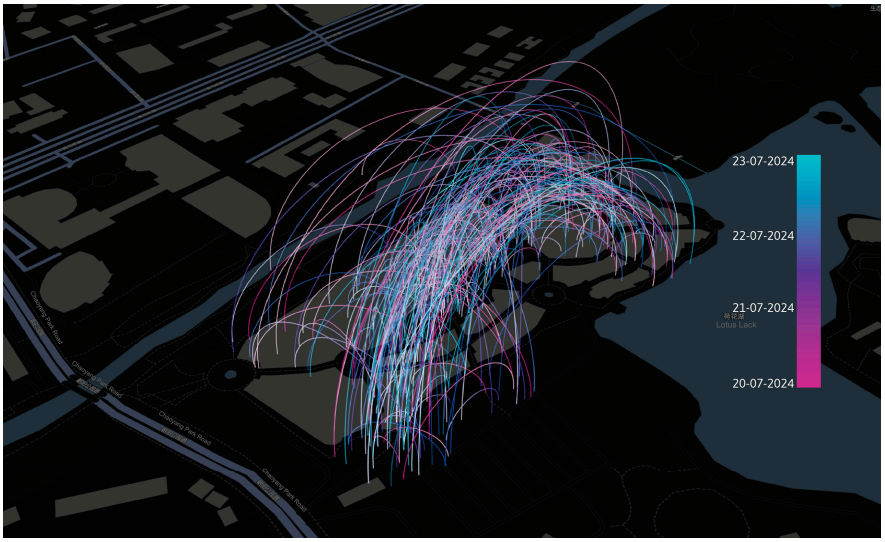


Figure 9. Gensen user check-in point tracing based on time, illustrating a temporary origin–destination relationship.

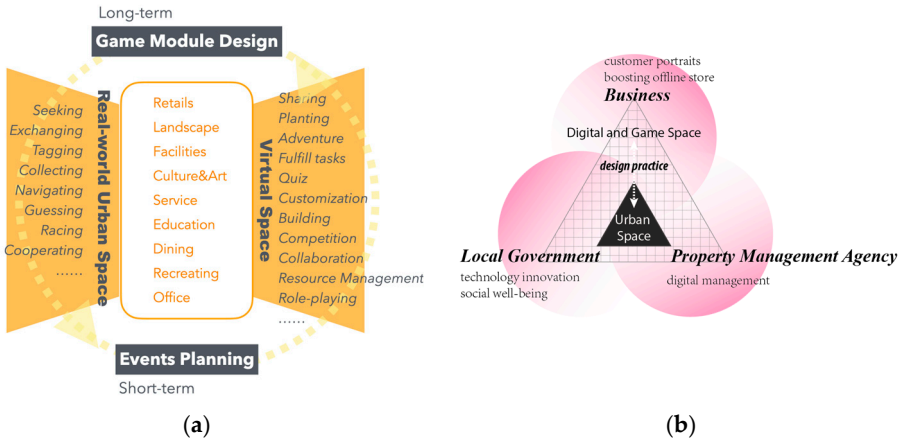
While anecdotal observations and spatial data imply increased activity in underutilized zones—such as indoor retail corridors and secondary entrances—these findings alone are insufficient to substantiate complex social outcomes such as "social mixing," "community building", or enhanced "accessibility." The check-in data captures movement and spatial reach but does not reveal the demographic diversity of participants, the nature of their interactions, or whether meaningful social bonds were formed. Therefore, any claims regarding social sustainability impacts should be interpreted with caution.

Nonetheless, the observed changes in mobility patterns and engagement offer preliminary insight into how a gamified design approach may facilitate conditions conducive to social sustainability. By transforming urban spaces into interactive platforms, Gensen introduces a playful logic that encourages exploration and shared participation. The hybrid digital/physical nature of the event allows for place-based interaction without disrupting the commercial or functional operations of the site.

To advance from concept to measurable impact, a gameful design approach must be developed through a co-design strategy that integrates online platforms with offline spatial contexts. As illustrated in Figure 10a, this involves positioning existing urban functions as interfaces for game modules—where virtual objectives align with real-world activities. For example, Gensen's "Urban Gardening" module complements on-site programming like seasonal flower shows and art festivals, offering an opportunity to create continuity between virtual play and physical event-making.

Institutional collaboration remains a critical factor. The SOLANA pilot benefited from cooperation between Gensen, property managers, and local commercial stakeholders. Future efforts must scale this model through inclusive governance structures that involve municipal authorities, developers, and community groups (Figure 10b). Additionally, digital tools integrated into urban management systems—such as for security, energy efficiency, or resource analytics—can support more responsive planning.

However, the challenge persists in balancing institutional objectives with the lived experiences and needs of citizens. Without triangulation from qualitative data (e.g., user interviews, ethnographic observations) and attention to equity and representation, gamified interventions risk becoming short-term spectacles rather than long-term strategies for social transformation. Future studies should explore these dimensions more rigorously to assess the true potential of gameful urban design in promoting social sustainability.



(**a**) (**b**)

Figure 10. Framework for a game-aided urban design approach can be further explored by incorporating games familiar to users, while also introducing the main mechanics of these games (**a**). Collaboration among stakeholders plays a critical role in deploying the approach, which should run in parallel with the design process (**b**).

5.2. Rethinking Urban Gamification: Comparing Gensen's Task-Oriented Play with Foursquare's Identity-Driven Engagement

The Treasure Hunting activity in Gensen can be recognized as a familiar game structure, widely understood as a task involving the search for hidden items or goals. This format has been applied across multiple domains to enhance engagement, such as in retail environments to drive customer interaction [59], in language learning to improve motivation and vocabulary retention [60], and in marketing to influence purchase intention and brand loyalty in metaverse contexts [61]. In the context of the Gensen platform, Treasure Hunting incorporates several well-established gameful design elements—including goal-oriented tasks, spatial exploration, challenges, progress tracking, and a reward system. These components align with Deterding's [18] definition of gamification as "the use of game design elements in non-game contexts," applied here within a real urban setting.

While the activity effectively integrates game mechanics into the urban environment, it overlooks the inherent playfulness already embedded in physical space and human interaction. This critique echoes Bogost's [62] argument that play is not merely a layer imposed onto a setting but a mode of engagement with the limitations and affordances of space itself. From this perspective, Treasure Hunting in Gensen risks instrumentalizing space through gamification rather than fostering emergent play rooted in the material and cultural qualities of the environment. It is therefore critical to understand the unique characteristics of local areas before implementing gamified elements. Introducing "common-sense" or standardized games without sensitivity to local culture may result in spatial and experiential homogenization across different districts of a city. This effect, when scaled, could pose a threat to cultural diversity by erasing localized forms of social interaction and identity. In turn, such homogenization undermines social sustainability by weakening the sense of place, belonging, and community cohesion—especially in culturally rich or diverse urban contexts.

Furthermore, Frith's [63] interviews with Foursquare users reveal that gamified elements—particularly badges and mayorships—directly influenced mobility decisions, even if the behavioral changes appeared minor. These decisions were not just about movement but also about how places were valued, often in competitive or identity-affirming ways. In contrast, Gensen's gamification of urban space is more event-driven and goal-oriented, compelling users to visit specific locations to earn rewards (e.g., a coffee voucher). While Gensen successfully increases mobility across underutilized zones in SOLANA (e.g.,

the eastern indoor areas or secondary entrances), it lacks the deeper identity construction elements seen in Foursquare.

Frith argues that Foursquare's gamification allows users to perform identity through consumption and spatial practices, such as collecting culturally meaningful venues (e.g., libraries, pho restaurants). In contrast, Gensen does not yet offer such a nuanced layer of identity performance. The game is more about completing tasks within a pre-scripted event structure than curating a public self. This limits the platform's potential for long-term user retention, personalization, or deeper social engagement, thereby restricting how gamified design could support social sustainability in a digitalized environment.

However, Gensen does offer a more direct mirroring of real-world activities in a virtual environment, creating a closer behavioral correspondence between digital and physical spaces. This interface could support the formation of virtual interest-based groups that may later evolve into location-based communities in reality. Furthermore, the platform's close collaboration with commercial spaces allows for stronger integration between the digital game environment and real-world experiences. Personal identity and neighborhood affiliation could be more effectively developed if Gensen were expanded beyond commercial zones and implemented more broadly across city districts. By increasing the complexity and dynamism of the virtual environment, user identity could be shaped through a richer set of behaviors and spatial choices. Additionally, enabling users to initiate their own events based on personal interests would further enhance identity expression and support grassroots community formation.

5.3. Design for Social Sustainability in Digitalized Urban Contexts

Sustainability is a context-dependent concept [64] that has also been embraced by urban planning and design in response to the continuously growing urban population. In the urban context, social sustainability is often viewed from a functional perspective, focusing on the long-term survival of viable urban social units, promoting social cohesion, and increasing accessibility to public spaces and jobs [65]. The form of urban space is considered vital for social sustainability development and plays a crucial role in shaping social relations and cultural practices [66,67]. A digitally innovative urban space, distinct from traditional material spaces, requires a rethinking of spatial typology in terms of function, form, and structure. This need arises not only from the influence of digital technology on space but also from the evolving everyday activities of individuals. A new spatial culture is emerging due to the changing tempo-spatial distribution of activities.

The shift toward digitalized lifestyles and spaces in cities, driven by rapid advancements in technology and the integration of digital tools in everyday life, has significant implications for spatial disparity. As digital innovation reshapes urban environments, it introduces new forms of inequality, particularly concerning access to digital spaces and the ability to participate in a digitalized lifestyle. Importantly, disparities in digital literacy also affect who can meaningfully engage with gamified urban systems, potentially excluding elderly users, individuals with limited education, or those unfamiliar with emerging technologies. Despite these challenges, gamification presents a promising avenue to promote social sustainability in digitalized cities. Gamified experiences can be designed to be intuitive and accessible, allowing people with varying levels of digital literacy to engage. Moreover, gamified urban environments can be designed to foster community building and social mixing by creating shared goals and challenges that bring diverse groups together.

Therefore, Gensen's pilot Treasure Hunting event presents a valuable opportunity to further explore how the gamification of urban space—when situated within an advanced digitalized environment—can contribute meaningfully to social sustainability. However, the analysis also reveals critical limitations that warrant attention. First, Gensen's current

top-down, event-driven structure restricts long-term user engagement and undermines its potential for sustained social relevance. A transition toward participatory and inclusive design—particularly involving underrepresented and marginalized communities—is essential for enhancing the cultural resonance and equity of such gamified interventions. Second, while the platform effectively activates underutilized areas of the urban fabric, its reliance on smartphone-based interaction may exclude those lacking adequate access or digital literacy. To address this, Gensen should adopt hybrid mechanisms, such as physical game markers, printed materials, or public displays, to expand accessibility and foster broader community participation. Finally, the platform's current commercial orientation must be critically reconsidered. To more fully support social sustainability, gamification should be redirected toward civic and environmental objectives—for instance, urban gardening, local storytelling, or neighborhood clean-up challenges. Such a purpose-driven approach would enable deeper community building and align gamified urban design more closely with inclusive, socially responsive outcomes.

6. Conclusions

This paper explored how gameful design approaches can support social sustainability in rapidly evolving urban contexts. Specifically, it addressed the question: How can gameful design approaches be deployed within existing urban fabrics and local cultures to foster social interaction and engagement with space?

Using the Gensen platform's implementation in the SOLANA commercial area of Beijing as a case study, we examined how digital gamified experiences can shape individual behavioral patterns and transform engagement with long-standing urban spaces. By enhancing spatial accessibility and promoting dynamic interactions, Gensen demonstrates the potential of integrating virtual environments with physical urban form. Its cooperation model—linking business stakeholders, local government, and property management—also offers institutional support for implementation. However, while Gensen shows promise in facilitating co-presence and forming new cultural communities, its current top-down design limits more spontaneous or personalized play. The dominant use of a fixed game type (e.g., treasure hunting) restricts user identity expression and grassroots community formation. By comparing Gensen with Foursquare, this study highlights the importance of individual identity construction in gamified urban experiences. Features such as user-generated events—already embedded in Gensen—could be leveraged to enhance personalization, allowing identity-driven engagement and organic community development.

This study emphasizes that achieving social sustainability through gamification requires balancing top-down planning with bottom-up participation. Practical tools such as co-design workshops and user-initiated in-app events could empower users to shape their environment and strengthen local attachment. Moreover, optimizing Gensen's data collection functions could increase public awareness and knowledge about local spaces, further supporting participatory design and community empowerment.

This research fills a key gap in the literature by showing how gameful design, when situated within real urban contexts, can impact socially sustainable development. It contributes original insights into how digital platforms like Gensen can reimagine urban space by bridging virtual interventions with lived experience. Ultimately, the findings underscore the need for more inclusive, identity-sensitive, and participatory strategies in the gamification of cities—approaches that can foster socially cohesive, vibrant urban communities.

Author Contributions: Conceptualization, Y.Y.; methodology, Y.Y. and Z.L.; validation, Y.Y., Z.L. and S.M.; formal analysis, Y.Y.; investigation, Y.Y.; data curation, Y.Y. and Z.L.; writing—original draft preparation, Y.Y.; writing—review and editing, Y.Y., Z.L. and S.M.; visualization, Y.Y.; supervi-

sion, S.M.; funding acquisition, Z.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Conflicts of Interest: Author Zheyu Li was employed by the company Prismo Studio. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. McKenzie, S. *Social Sustainability: Towards Some Definitions*; Hawke Research Institute Working Paper Series No 27; Hawke Research Institute: Magill, Australia, 2004.
2. Dempsey, N.; Bramley, G.; Power, S.; Brown, C. The Social Dimension of Sustainable Development: Defining Urban Social Sustainability. *Sustain. Dev.* **2011**, *19*, 289–300. [CrossRef]
3. Vallance, S.; Perkins, H.C.; Dixon, J.E. What is social sustainability? A clarification of concepts. *Geoforum* **2011**, *42*, 342–348. [CrossRef]
4. Eizenberg, E.; Yosef, J. Social sustainability: A new conceptual framework. *Sustainability* **2017**, *9*, 68. [CrossRef]
5. Woodcraft, S. Understanding and measuring social sustainability. *J. Urban Regen. Renew.* **2015**, *8*, 133. [CrossRef]
6. Manzi, T.; Lucas, K.; Lloyd, T.J.; Allen, J. *Social Sustainability in Urban Areas: Communities, Connectivity and the Urban Fabric*; Routledge: New York, NY, USA, 2010.
7. Chan, E.; Lee, G.K.L. Critical factors for improving social sustainability of urban renewal projects. *Soc. Indic. Res.* **2008**, *85*, 243–256. [CrossRef]
8. Bramley, G.; Dempsey, N.; Power, S.; Brown, C. What is 'social sustainability', and how do our existing urban forms perform in nurturing it. In Proceedings of the Sustainable Communities and Green Futures' Conference, London, UK, 5–7 April 2006.
9. Baek, J.; Choe, Y. Differential effects of the valence and volume of online reviews on customer share of visits: The case of US casual dining restaurant brands. *Sustainability* **2020**, *12*, 5408. [CrossRef]
10. Glaeser, E.L.; Kominers, S.D.; Luca, M.; Naik, N. Big data and big cities: The promises and limitations of improved measures of urban life. *Econ. Inq.* **2016**, *56*, 114–137. [CrossRef]
11. Kamrowska-Załuska, D. Impact of AI-based tools and urban big data analytics on the design and planning of cities. *Land* **2021**, *10*, 1209. [CrossRef]
12. Quan, S.J.; Park, J.; Economou, A.; Lee, S. Artificial intelligence-aided design: Smart Design for sustainable city development. *Environ. Plan. B Urban Anal. City Sci.* **2019**, *46*, 1581–1599. [CrossRef]
13. Lee, C.-S.; Wong, K.-S.D. Deriving a gamified learning-design framework towards sustainable community engagement and mashable innovations in Smart Cities: Preliminary findings. *Int. J. Knowl. Syst. Sci. (IJKSS)* **2018**, *9*, 22. [CrossRef]
14. Hamari, J.; Koivisto, J.; Sarsa, H. Does gamification work?—A literature review of empirical studies on gamification. In Proceedings of the 47th Annual Hawaii International Conference on System Sciences, Waikoloa, HI, USA, 6–9 January 2014; IEEE: Piscataway, NJ, USA, 2014.
15. Deterding, S.; Sicart, M.; Nacke, L.; O'Hara, K.; Dixon, D. Gamification. using game-design elements in non-gaming contexts. In Proceedings of the CHI'11 Extended Abstracts on Human Factors in Computing Systems. Vancouver, BC, Canada, 7–12 May 2011; pp. 2425–2428.
16. El Shoubashy, H.; El Kader, H.A.; Khalifa, N. What is gamification? A literature review of previous studies on gamification. *Aust. J. Basic Appl. Sci.* **2020**, *14*, 29–51.
17. Attali, Y.; Meirav, Y.A.A. Gamification in assessment: Do points affect test performance? *Comput. Educ.* **2015**, *83*, 57–63. [CrossRef]
18. Deterding, S.; Dixon, D.; Khaled, R.; Nacke, L. From game design elements to gamefulness: Defining "gamification". In Proceedings of the MindTrek '11: Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments, Tampere, Finland, 29–30 September 2011; pp. 9–15.
19. Huotari, K.; Hamari, J. A definition for gamification: Anchoring gamification in the service marketing literature. *Electron. Mark.* **2017**, *27*, 21–31. [CrossRef]
20. Fortes Tondello, G.; Mora, A.; Nacke, L. Elements of gameful design emerging from user preferences. In Proceedings of the Annual Symposium on Computer-Human Interaction in Play, Amsterdam, The Netherlands, 15–18 October 2017.
21. Fortes Tondello, G.; Nacke, L.E. Towards Customizing Gameful Systems by Gameful Design Elements. In Proceedings of the Personalization in Persuasive Technology Workshop, Persuasive Technology, Waterloo, ON, Canada, 18–19 April 2018.

22. Landers, R.N.; Tondello, G.F.; Kappen, D.L.; Collmus, A.B.; Mekler, E.D.; Nacke, L.E. Defining gameful experience as a psychological state caused by gameplay: Replacing the term 'Gamefulness' with three distinct constructs. *Int. J. Hum.-Comput. Stud.* **2019**, *127*, 81–94. [CrossRef]
23. Sailer, M.; Hense, J.U.; Mayr, S.K.; Mandl, H. How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Comput. Hum. Behav.* **2017**, *69*, 371–380. [CrossRef]
24. Holden, J.I.; Kupperman, J.; Dorfman, A.; Saunders, T.; Amanda, P.; Pagan, M. Gameful learning as a way of being. *Int. J. Learn. Technol.* **2014**, *9*, 181–201. [CrossRef]
25. Ioannou, A. A model of gameful design for learning using interactive tabletops: Enactment and evaluation in the socio-emotional education classroom. *Educ. Technol. Res. Dev.* **2019**, *67*, 277–302. [CrossRef]
26. Budde, M.; Öxler, R.; Beigl, M.; Holopainen, J. Sensified gaming: Design patterns and game design elements for gameful environmental sensing. In Proceedings of the 13th International Conference on Advances in Computer Entertainment Technology, Osaka, Japan, 9–12 November 2016; Volume 3, pp. 1–8.
27. Hoffmann, G.; Pfeiffer, J. Gameful learning for a more sustainable world: Measuring the effect of design elements on long-term learning outcomes in correct waste sorting. *Bus. Inf. Syst. Eng.* **2022**, *64*, 459–482. [CrossRef]
28. Woetzel, J.; Remes, J.; Boland, B.; Lv, K.; Sinha, S.; Strube, G.; Means, J.; Law, J.; Cadena, A.; von der Tann, V. *Executive Summary, Smart Citiess: Digital Solution for a More Livable Future*. 2018. Available online: https://www.mckinsey.com/capabilities/operations/our-insights/smart-cities-digital-solutions-for-a-more-livable-future (accessed on 23 April 2025).
29. Zhang, J.; He, S. Smart technologies and urban life: A behavioral and social perspective. *Sustain. Cities Soc.* **2020**, *63*, 102460. [CrossRef]
30. Angela Pearce, M.S. The digital gap still exists, generationally, rurally, and academically. In Proceedings of the 12th International Conference on Education Technology and Computers, London, UK, 23–26 October 2020.
31. Allam, Z.; Newman, P. Redefining the Smart City: Culture, Metabolism and Governance. *Smart Cities* **2018**, *1*, 4–25. [CrossRef]
32. Atkinson, R.; Willis, P. Transparent cities: Re-shaping the urban experience through interactive video game simulation. *City* **2009**, *13*, 403–417. [CrossRef]
33. Shen, C.; Monge, P.; Williams, D. Virtual brokerage and closure: Network structure and social capital in a massively multiplayer online game. *Commun. Res.* **2014**, *41*, 459–480. [CrossRef]
34. Szell, M.; Thurner, S. Measuring social dynamics in a massive multiplayer online game. *Soc. Netw.* **2010**, *32*, 313–329. [CrossRef]
35. Ganley, D.; Lampe, C. The ties that bind: Social network principles in online communities. *Decis. Support Syst.* **2009**, *47*, 266–274. [CrossRef]
36. Torres, E.N. Online-to-offline interactions and online community life cycles: A longitudinal study of shared leisure activities. *Leis. Sci.* **2022**, *42*, 32–50. [CrossRef]
37. Guthrie, C.; Fosso-Wamba, S.; Arnaud, J.B. Online consumer resilience during a pandemic: An exploratory study of e-commerce behavior before, during and after a COVID-19 lockdown. *J. Retail. Consum. Serv.* **2021**, *61*, 102570. [CrossRef]
38. Kawasaki, T.; Wakashima, H.; Shibasaki, R. The use of e-commerce and the COVID-19 outbreak: A panel data analysis in Japan. *Transp. Policy* **2022**, *115*, 88–100. [CrossRef]
39. Gao, X.; Shi, X.; Guo, H.; Liu, Y. To buy or not buy food online: The impact of the COVID-19 epidemic on the adoption of e-commerce in China. *PLoS ONE* **2020**, *15*, e0237900. [CrossRef] [PubMed]
40. Huseynov, F. Gamification in e-commerce: Enhancing digital customer engagement through game elements. In *Digital Innovations for Customer Engagement, Management, and Organizational Improvement*; IGI Global: Hershey, PA, USA, 2020; pp. 144–161.
41. Arifah, I.D.C. E-Commerce Gamification: The Effect of gameful Experience (Gamex) and Game Design on The Self-brand Connection. In Proceedings of the 3rd International Conference on Business and Management of Technology (ICONBMT 2021), Online, 31 July 2021; Atlantis Press: Dordrecht, The Netherlands, 2021.
42. Grüger, D.; Weiblen, J.; Weber, P.; Ludwig, T. How Gamified Are Sustainable Food Apps? Applying the Gameful Design Heuristics to Evaluate Sustainable Food Apps. *Proc. ACM Hum.-Comput. Interact.* **2023**, *7*, 919–947. [CrossRef]
43. Panattil, S.J.; George, A.; Sajid, M.; Joy, M.M. Driving affective commitment among food delivery executives towards an online food delivery partner platform through gamification. *J. Foodserv. Bus. Res.* **2023**, 1–31. [CrossRef]
44. Costa Bomfim, M. *A Recipe for Food Literacy: Designing and Evaluating Technologies for Informed Food Choices*; University of Waterloo: Waterloo, ON, Canada, 2022.
45. Allam, Z.; Sharifi, A.; Bibri, S.E.; Jones, D.S.; Krogstie, J. The metaverse as a virtual form of smart cities: Opportunities and challenges for environmental. *Smart Cities* **2022**, *5*, 771–801. [CrossRef]
46. De Souza e Silva, A.; Hjorth, L. Playful urban spaces: A historical approach to mobile games. *Simul. Gaming* **2009**, *40*, 602–625. [CrossRef]
47. Sharpe, E.K.; Glover, T.D. *Placemaking in the Playful City: Playing in and Playing with the Urban Environment*; Leisure Communities; Routledge: London, UK, 2020; pp. 91–99.

48. Donoff, G.; Bridgman, R. The playful city: Constructing a typology for urban design interventions. *Int. J. Play.* **2017**, *6*, 294–307. [CrossRef]
49. Boyer, M.C. *The City of Collective Memory: Its Historical Imagery and Architectural Entertainments*; Mit Press: Cambridge, MA, USA, 1994.
50. Alfrink, K. *The Gameful City*; Walz, S.P., Deterding, S., Eds.; Mit Press: Cambridge, MA, USA, 2015.
51. Sanoff, H. *Community Participation Methods in Design and Planning*; John Wiley & Sons: Toronto, ON, Canada, 1999.
52. De Lange, M. The playful city: Using play and games to foster citizen participation. In *Social Technologies and Collective Intelligence*; Skaržauskaitė, M., Skarzauskiene, A., Ewart, J., Rynkiewicz, B., Eds.; Mykolas Romeris University: Vilnius, Lithuania, 2015; pp. 426–434.
53. Ampatzidou, C.; Vervoort, J.; von Flittner, Z.F.; Vaajakallio, K. New Insights, New Rules: What Shapes the Iterative Design of an Urban Planning Game? *Urban Plan.* **2022**, *7*, 295–305. [CrossRef]
54. Angelidou, M.; Psaltoglou, A. Social innovation, games and urban planning: An analysis of current approaches. *Int. J. Electron. Gov.* **2019**, *11*, 5–22. [CrossRef]
55. Ferri, F.; Grifoni, P.; Guzzo, T. Online Learning and Emergency Remote Teaching: Opportunities and Challenges in Emergency Situations. *Societies* **2020**, *10*, 86. [CrossRef]
56. Fox, N.; Campbell-Arvai, V.; Lindquist, M.; Van Berkel, D.; Serrano-Vergel, R. Gamifying decision support systems to promote inclusive and engaged urban resilience planning. *Urban Plan.* **2022**, *7*, 239–252. [CrossRef]
57. Ma, D.; Ma, Y. The impact of the digital economy on quality of life: Empirical evidence from 220 cities in China. *Digit. Policy, Regul. Gov.* **2025**, *27*, 404–421. [CrossRef]
58. Zhang, R.; Yao, G.; Luo, Y. Research on Small-Scale Characteristics of Urban Vitality Space Driven by Multi-Source Sentiment Data: With "Xidan The New" and "Beijing Fun" in Beijing as Examples. *China City Plan. Rev.* **2024**, *33*, 44–54.
59. Olsson, M.; Hogberg, J.; Wastlund, E.; Gustafsson, A. In-store gamification: Testing a location-based treasure hunt app in a real retailing environment. In Proceedings of the 2016 49th Hawaii International Conference on System Sciences (HICSS), Koloa, HI, USA, 5–8 January 2016; IEEE: Piscataway, NJ, USA, 2016.
60. Hussin, S.; Abdullah, M.A.R. The Impact of Treasure Hunt Games in Enhancing Basic Japanese Vocabulary Retention and Student Motivation through Gamified Learning. *Int. J. Acad. Res. Bus. Soc. Sci.* **2024**, *14*, 2222–6990. [CrossRef]
61. Al-Adwan, A.S.; Yaseen, H.; Alkhwaldi, A.F.; Jafar, R.M.S.; Fauzi, M.A.; Abdullah, A. Treasure hunting for brands: Metaverse marketing gamification effects on purchase intention, WOM, and loyalty. *J. Glob. Mark.* **2025**, 1–25. [CrossRef]
62. Bogost, I. *Play Anything: The Pleasure of Limits, the Uses of Boredom, and the Secret of Games*; Basic Books: New York, NY, USA, 2016.
63. Frith, J. Turning life into a game: Foursquare, gamification, and personal mobility. *Mob. Media Commun.* **2013**, *1*, 248–262. [CrossRef]
64. Maloutas, T. Promoting social sustainability The case of Athens. *City* **2003**, *7*, 167–181. [CrossRef]
65. Polese, M.; Stren, R.E. (Eds.) *The Social Sustainability of Cities: Diversity and the Management of Change*; University of Toronto Press: Toronto, ON, Canada, 2000.
66. Tonkiss, F. *Cities by Design: The Social Life of Urban Form*; John Wiley & Sons: Cambridge, UK, 2014.
67. Kirby, A. The production of private space and its implications for urban social relations. *Politi-Geogr.* **2008**, *27*, 74–95. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.